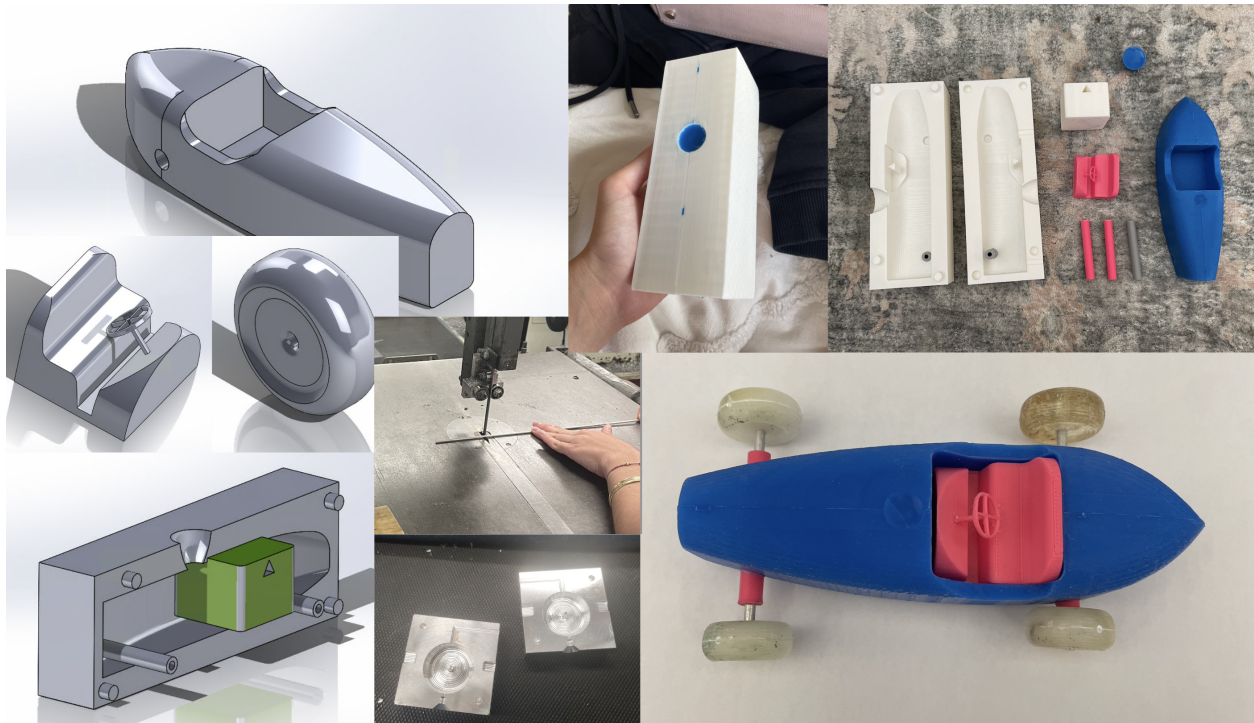


Small Scale Manufacturing of Bugatti Type 51 Toy Car

Team 17

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Abstract

The goal of this project was to create a small-scale model of a 1920s race car. The miniature replica we built, which consists of a car body, cockpit, wheels, and axles, gave us the tools to apply our class knowledge to a real world scenario. This project taught us how to practice the various manufacturing methods learned in class, such as machining, SOLIDWORKS, FUSION360 for computer aided manufacturing (CAM), additive manufacturing, and injection molding.

1 Manufacturing Process Overview

1.1 CAD Modeling and Design

To design a toy version of the 1931 Bugatti Type 51, we began by researching the blueprints. We researched the sizes of each component of the car. From those given dimensions, actually modeling the car on CAD

became a mix of art and creative problem solving. Using SOLIDWORKS, our group digitally modeled a simplified, scaled four-part assembly of the car, consisting of the car body, the cockpit, the axle holders, and the wheels. We did so by calculating the scaled dimensions, and designing the car body to follow the appropriate shape. The following components were modeled in reference to that scale, as well as the given constraints, such as for the wheel to fit $\frac{1}{8}$ " axle rods and the external dimensions to fit the available stock material for the wheel mold. Overall, the process required repeated refinements and adjustments to balance appearance, manufacturability, and dimensional accuracy throughout the CAD design stage.

1.2 Car Body: Mold Fabrication and Silicone Casting

Using SOLIDWORKS, we first created a spacer for the cockpit to fit into once the casting was done. This was done by extrude cutting a thin filleted rectangle onto the car body. Then, we made a negative of the designed car body into a rectangular box with the appropriate dimensions, given as a design constraint. We split it along the parting line (vertically through the car), and modeled a simple sprue and vent system for the silicone pouring and casting process. In addition, we incorporated the axle holders into the mold, and added alignment pins to ensure the mold was properly oriented. The components of the mold were then 3D printed, and assembled for casting. Our initial design did not account for enough clearance given the quality of PLA printing, so we had to reprint the mold to fit the axles into the mold cavities. Unfortunately, the original axles still broke during removal, but this did not impede the casting of the car, nor affect the quality of the axle holes on the car. We simply reprinted the axle holders to match our hot-pink cockpit.

1.3 Cockpit

The cockpit was built around the base dimensions of the spacer that was designed for the car body mold. We designed the wheel to be as aesthetically similar to the original, as well as the simplified shape of the seat cushion. 3D Printing with PLA required supports, which posed a challenge as the wheel architecture was quite thin. This was not removed as gently as it perhaps should have, which resulted in the wheel snapping off, and an emergency gorilla glue restoration, which worked quite well.

1.4 Wheel Mold: CAM & Machining

The wheel mold was made using Fusion 360 CAM, where we took our CAD design and turned it into toolpaths for machining two aluminum mold halves. We started by setting up the stock and aligning the coordinate system so everything would match correctly on the machine. From there, the machining was done in stages: first a facing pass to flatten the surface, then a roughing operation (adaptive clearing) to remove most of the material, and finally finishing passes to shape the wheel cavity and smooth it out. We also added drilling operations for the alignment pin holes and nozzle feature. Before cutting anything, we ran simulations to make sure there were no collisions and that all material would be removed properly. The molds were then machined on a Tormach mill, where careful setup was important to keep both halves aligned. Along the way, we made small adjustments to speeds and step sizes to get a better surface finish, especially in the curved areas. Overall, this process required a mix of planning and iteration to end up with a mold that was both accurate and usable for injection molding.

1.5 Wheels: Injection Molding

Injection molding is a process which is used to make parts by melting plastic and forcing it into a mold, where it cools and takes shape. After finishing the machining of the wheels, we wanted to mold them so that we would be able to then attach them to our car. One important factor we had to account for was plastic shrinkage, as the plastic slightly contracts once it cools. To address this problem, we designed the wheel mold to be slightly larger than the given dimensions so that the final wheels would match their intended size after cooling. For the molding process, we loaded plastic pellets into a machine at a temperature of approximately 280 K. The plastic from the machine was then injected into the mold and after it cooled, the wheels were removed, while any small excess material was manually trimmed.

1.6 Assembly and Finishing

After all the components were manufactured, the final step was to assemble everything together. During the casting process, a small amount of excess silicone, known as flash, formed along the seams where the two mold halves met. This excess material was carefully removed by hand, while thicker pieces left from the gating system were trimmed using a sharp knife to create cleaner edges and improve the overall fit of the parts. One of the main challenges during assembly was fitting the wheels onto the metal axle rods. Due to slight tolerancing differences and small imperfections from the injection molding process, some wheels fit very tightly on the axle and required careful adjustment to avoid damaging the parts. Sandpaper was also used to slightly smooth and flatten sections of the axle rods, helping the wheels slide on more evenly and improving rotation. Once properly aligned, the axles were inserted through the axle holders molded into the car body, allowing the wheels to rotate while remaining securely attached. Finally, the 3D-printed cockpit was fitted into the body and glued in place, completing the final assembly of the model car.

2 Project Reflections and Analysis

2.1 What Went Well

Creating the base CAD models went fairly well, and allowed for easy collaboration. Despite needing a second iteration, the 3D printing stage was also straightforward and provided predictable time-frames. Moreover, we were able to do the casting in one try, once the second mold fit our requirements. We also polished the axle edges after cutting them to length, which allowed for a smooth, not sharp, finish, and easier insertion into the PLA axle holders.

2.2 Challenges Faced

We encountered several challenges. For instance, we addressed issues with tolerancing, which arose with the mold when doing the silicone casting. This could have also been made easier with a better choice in 3D print surface finish quality to minimize friction between mating parts (spacer, axles, mold). For the injection molding, it took multiple consistent attempts to get the plastic to fully fill the wheel mold. A major challenge we faced was that the molten plastic would not enter the cavity because the injection nozzle was not perfectly centered with the mold opening. To fix this, we had to carefully adjust the position of the mold relative to the injection nozzle to ensure proper alignment before the injection. It took multiple tries until we finally were able to get the plastic to flow directly into the cavity and fully fill the mold. We also had to ensure that the plastic was fully melted and the machine was set to the appropriate temperature. Controlling these factors improved the quality of the wheels significantly.

2.3 Potential Improvements

A future iteration of this project would benefit from additional planning, such as initial tolerancing for 3D printed material and draft angles in the CNC wheel mold. Surface finishing could be improved by choosing a better filament surface quality when 3D printing the mold. Similarly, polishing the wheel mold could provide better surface quality, which would make removal of the wheels faster.

2.4 Final Reflection

This project provided essential hands-on experience in a complete manufacturing workflow, emphasizing the critical role of problem-solving and continuous iteration. Translating the Bugatti model from CAD to a physical assembly required ongoing refinement—including reprinting molds, modifying clearances, and adjusting injection alignments—to overcome real-world manufacturing challenges. Navigating these setbacks strengthened our ability to identify manufacturing defects, adapt our machining and casting processes, and successfully execute a multi-step engineering design.